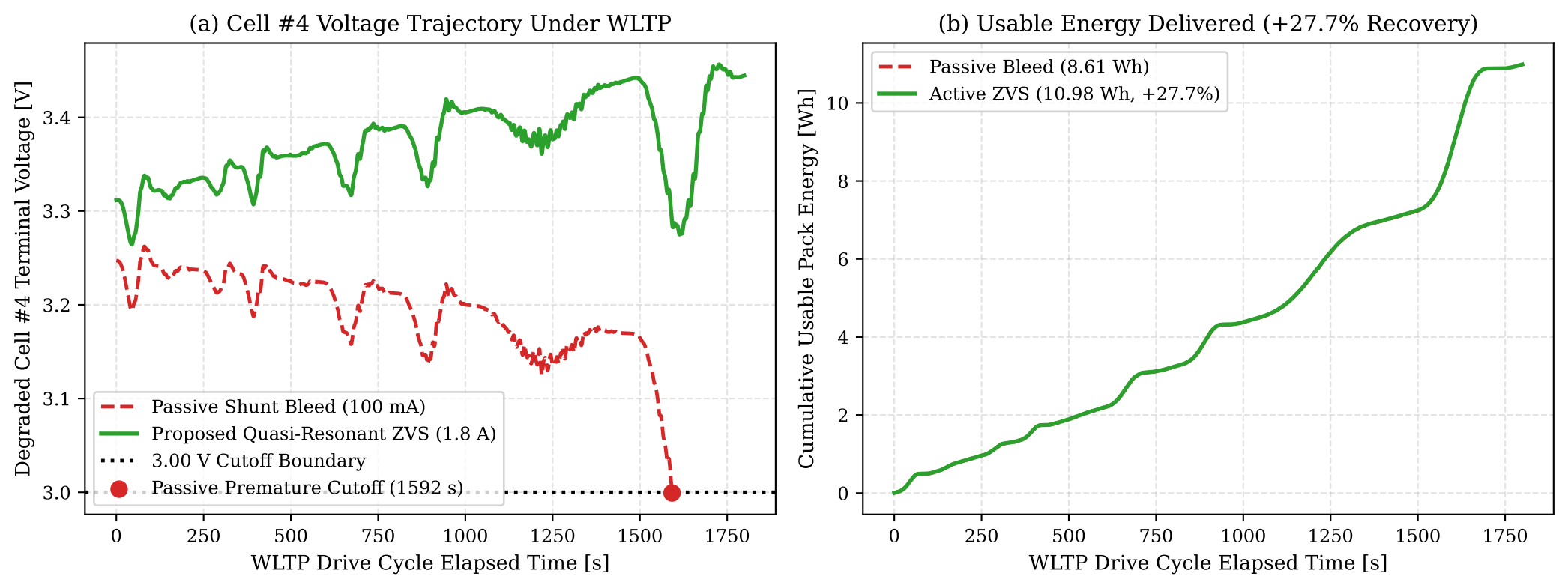
(a) Cell #4 Voltage Trajectory Under WLTP
Degraded Cell #4 Terminal Voltage [V]
3.4
3.3
3.2
3.1
3.0
Passive Shunt Bleed (100 mA)
Proposed Quasi-Resonant ZVS (1.8 A)
3.00 V Cutoff Boundary
Passive Premature Cutoff (1592 s)
0
250
500
750
1000
1250
1500
1750
WLTP Drive Cycle Elapsed Time [s]
(b) Usable Energy Delivered (+27.7% Recovery)
Cumulative Usable Pack Energy [Wh]
10
8
6
4
2
0
Passive Bleed (8.61 Wh)
Active ZVS (10.98 Wh, +27.7%)
0
250
500
750
1000
1250
1500
1750
WLTP Drive Cycle Elapsed Time [s]